

## Builder Basics — The Trail Maker (Extension · Intermediate)

**Topic:** Place On Move · **Course:** Builder Basics · **Type:** Extension Activity · **Level:** Intermediate · **Time:** about 38 minutes

 This is a bonus challenge, not a main lesson. Try it once you are comfortable placing blocks and you want the agent to place a whole line for you.

In the Placing Blocks lesson you placed blocks one at a time. Here, type `run` and the agent flips a switch called **place on move**, then walks forward — no code to type.

When **place on move** is **ON**, the agent drops a block on every step — a **trail**. When it is **OFF**, it walks but places nothing — a **gap**.

### 1 · Predict

Here is the order the program runs:

```
place on move ON
move forward by 4
place on move OFF
move forward by 2
place on move ON
move forward by 4
```

One block drops per step while the switch is ON. How many blocks come from each move — `forward 4`, then `forward 2` (OFF), then `forward 4`? Write all three numbers.

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**Add them up. How many blocks are placed in total, and how wide is the gap in the middle?**  
**Show your working.**

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2 · Trace the Switch 

Walk through the program one move at a time. For each move command, write the switch state and how many blocks that stretch places.

| Move command   | Switch (ON / OFF) | Blocks placed |
|----------------|-------------------|---------------|
| move forward 4 |                   |               |
| move forward 2 |                   |               |
| move forward 4 |                   |               |

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The agent walked 10 steps but placed fewer than 10 blocks. Why? Which stretch is the reason?

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### 3 · Reason About the Switch

No code here — think it through.

**The trail comes out as one solid line with NO gap. Which line of the program went wrong? Explain in one or two sentences.**

Imagine the very first **place on move ON** was missing, so the switch stayed OFF the whole time. What would be on the ground after the agent finished walking? Explain why.

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#### 4 · Make It Yours

The switch decides WHEN the agent places, and the move numbers decide HOW MANY blocks and how wide the gap is.

**Design a trail that places exactly 10 blocks with a 3-wide gap in the middle. Write your ON / OFF / move sequence, line by line.**

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**One sentence: how did you pick your move numbers to get 10 blocks?**

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run

## one video

**1 · Your code.** Scroll through it. Say what each part does.

**2 · Your build.** Point the camera. Name the parts.

Fill the blanks:

Today I built \_\_\_\_\_.

I built it using this code: \_\_\_\_.

In this code I used \_\_\_\_\_.

The hardest part was \_\_\_\_\_.

That part was hard because \_\_\_\_\_.

The most fun part was \_\_\_\_\_.

Something new I learned was \_\_\_\_\_.

**Write your lines here, then say them in your video.**

**Submit** 

Send this worksheet + your video to teacher on KakaoTalk.